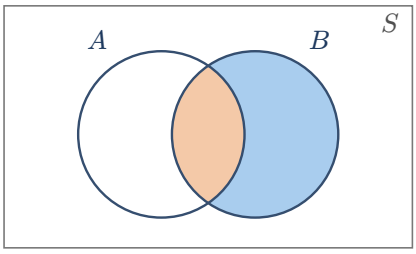
PROBABILITY AND STATISTICS I

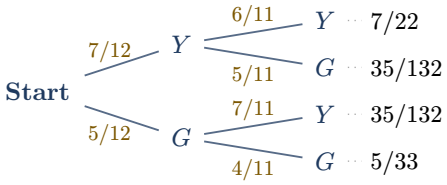
Conditional Probability

and Independence

Lecture 04

$P(A \mid B)$ · the multiplication rule · independent events · tree diagrams





Lecture notes with every worked example solved,
vector tree and Venn diagrams, the reasoning behind each formula,
and practice problems with answers.

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Previous chapter: Probability Axioms

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1 Conditional probability

Chapter 3 assigned a probability to each event once, before anything was known. But information arrives: the die has been rolled and someone tells you the number is even; the first card has been turned over and it is an Ace. Each piece of news rules some outcomes out, and the probabilities of what is left have to be updated. **Conditional probability** is the rule for that update.

Definition 4.1 — Conditional probability. The **conditional probability** of an event A given an event B, denoted $P(A \mid B)$, is defined as

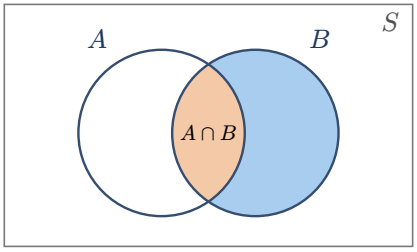
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad \text{if } P(B) \neq 0. \tag{I}$$

Similarly,

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \quad \text{if } P(A) \neq 0. \tag{II}$$

Read $P(A \mid B)$ as "the probability of A **given** B".

Why this formula. Once we know B has occurred, every outcome outside B is impossible: B becomes the new sample space. The only part of A that can still happen is the part inside B, namely $A \cap B$ (amber). So $P(A \mid B)$ is the share of B that $A \cap B$ takes up — its probability divided by the probability of the whole of B.



Dividing by $P(B)$ is a rescaling: it makes the new sample space B have probability 1, exactly as Axiom 2 demands of any sample space.

From Eqs. (I) and (II), multiplying out, we have

Rule 4.1 — Multiplication rule.

$$P(A \cap B) = P(A \mid B)\,P(B) = P(B \mid A)\,P(A).$$

The definition and the rule are the same equation, read two ways. When $P(A \cap B)$ is known, divide to get a conditional probability; when a conditional probability is easy to see directly — as it is in every "without replacement" problem below — multiply to get $P(A \cap B)$.

Example 1. In a one-time dice experiment, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$.

1. Find the probability of getting the number 2.
2. Find the probability of getting the number 2 if you know that the number that appeared is even.

Solution.

1. All six faces are equally likely, so $P(2) = 1/6$.
2. Knowing the number is even shrinks the sample space to $\{2, 4, 6\}$ — three equally likely outcomes, one of which is 2:

$$P(2 \mid \text{even}) = \frac{1}{3}.$$

The definition gives the same, with $A = \{2\}$ and $B = \{2, 4, 6\}$, so $A \cap B = \{2\}$:

$$P(2 \mid \text{even}) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{3/6} = \frac{1}{3}.$$

The information **changed** the probability, from 1/6 to 1/3. Hold on to that: in §3, events whose probabilities do **not** change are given a name.

Note. When outcomes are equally likely, the formula reduces to counting inside B:

$$P(A \mid B) = \frac{n(A \cap B)}{n(B)}.$$

So you can always either use the formula on the original sample space, or count directly in the reduced one. Example 1 did both; they always agree.

Note. For a fixed B, $P(\cdot \mid B)$ obeys all three axioms — it is a genuine probability on the new sample space. Every theorem of Chapter 3 therefore holds with "$\mid B$" added throughout; in particular

$$P(A^c \mid B) = 1 - P(A \mid B).$$

2 Drawing without replacement

Conditional probabilities are easiest to see in **sequential** experiments: draw one object, then another from what is left. The probability of the second draw depends on what the first one removed, and the condition "given the first was …" tells us exactly what was removed.

Example 2. A bag contains 3 red marbles and 4 blue marbles. Two marbles are drawn at random without replacement. If the first marble drawn is red, what is the probability the second marble is blue?

Solution. Let

$$A : \text{the first marble is red}, \quad B : \text{the second marble is blue}.$$

After the first marble is drawn and found to be red, there are now 6 marbles left in the bag, 4 of which are blue. Therefore

$$P(B \mid A) = \frac{4}{6} = \frac{2}{3}.$$

$$P(\text{the second marble is blue given that the first marble is red}) = \frac{2}{3}.$$

Note. **Check with the formula.** $P(A) = 3/7$, and the number of ordered draws with red first and blue second is $3 \times 4 = 12$ out of $7 \times 6 = 42$, so $P(A \cap B) = 12/42 = 2/7$. Then

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{2/7}{3/7} = \frac{2}{3}. \checkmark$$

Reasoning on the reduced bag is quicker — and it is the reason the formula is right.

Example 3. Two cards are chosen at random without replacement from a deck of 52 playing cards. If the first card chosen is an Ace, what is the probability the second card chosen is a King?

Solution. Let

$$A : \text{the first card is an Ace}, \quad B : \text{the second card is a King}.$$

After the first card is chosen and found to be an Ace, there are now 51 cards left in the deck, 4 of which are Kings — removing an Ace took no King away. Therefore

$$P(B \mid A) = \frac{4}{51}.$$

$$P(\text{the second card is a King given that the first card is an Ace}) = \frac{4}{51}.$$

Watch out. Change the question to "if the first card is a **King**, what is the probability the second is a King?" and the answer becomes 3/51: now the first draw **did** remove a King. Always ask what the condition took out of the bag.

Example 4. A box contains 5 green pencils and 7 yellow pencils. Two pencils are chosen at random from the box without replacement. What is the probability they are both yellow?

Solution. Let

$$A : \text{the first pencil is yellow}, \quad B : \text{the second pencil is yellow}.$$

Initially there are 12 pencils, 7 of which are yellow. If a yellow pencil is chosen, there will be 11 pencils left, 6 of which are yellow. So

$$P(A) = \frac{7}{12}, \quad P(B \mid A) = \frac{6}{11}.$$

"Both yellow" is $A \cap B$. Since

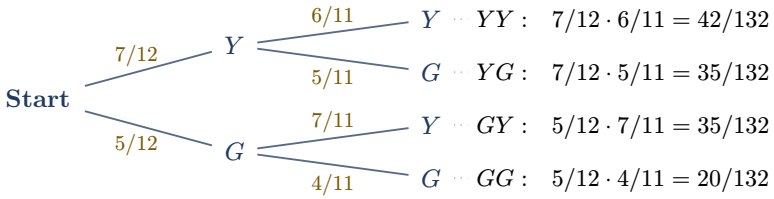
$$P(B \mid A) = \frac{P(A \cap B)}{P(A)},$$

the multiplication rule gives

$$P(A \cap B) = P(A) \times P(B \mid A) = \frac{7}{12} \times \frac{6}{11} = \frac{7}{22}.$$

2.1 Probability trees

A sequential experiment is drawn as a tree. Each branch is labelled with the probability of taking it **given everything to its left** — which is why the second-stage numbers are conditional. The probability of a path is the product along it: Rule 4.1, applied once per stage. Here is Example 4 in full (Y = yellow, G = green):



Note. Two free checks on any tree. The branches leaving each node sum to 1 ($7/12 + 5/12$, $6/11 + 5/11$, …), and the leaves sum to 1 ($42 + 35 + 35 + 20 = 132$). The tree also answers questions the example did not ask: "one of each colour" is the two middle leaves, $70/132 = 35/66$.

The rule extends to any number of stages. For three events,

$$P(A \cap B \cap C) = P(A)\,P(B \mid A)\,P(C \mid A \cap B),$$

each factor conditioned on everything before it — three Aces in a row from a deck is $4/52 \cdot 3/51 \cdot 2/50$.

Exercise set 1

1.1 A fair die is rolled once. Find $P(6 \mid \text{greater than } 3)$ and $P(\text{greater than } 3 \mid \text{even})$.

1.2 If $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, find $P(A \mid B)$, $P(B \mid A)$ and $P(A \mid B^c)$.

1.3 Two cards are drawn without replacement from a standard deck. Find the probability that both are hearts, and the probability that the first is a heart and the second a spade.

1.4 From the bag of Example 2 (3 red, 4 blue) two marbles are drawn without replacement. Draw the tree, and find the probability that both are red and that they are of different colours.

1.5 Two fair dice are rolled. Find the probability that the total is 8 given that the first die shows 3, and the probability that the first die shows 3 given that the total is 8.

1.6 Three cards are drawn without replacement. Find the probability that all three are Aces.

3 Independent events

Independent events are not affected by each other: knowing that one occurred does not change the probability of the other. For example, when tossing a coin repeatedly, heads or tails on the next toss is not affected by previous tosses — the coin has no memory.

In symbols, A is independent of B when conditioning on B changes nothing:

$$P(A \mid B) = P(A) \quad \text{or, equivalently,} \quad P(B \mid A) = P(B).$$

Substituting into the definition of conditional probability:

$$\because P(A \mid B) = \frac{P(A \cap B)}{P(B)} \text{ and } P(A \mid B) = P(A) \quad \therefore \frac{P(A \cap B)}{P(B)} = P(A) \quad \therefore P(A \cap B) = P(A)\,P(B).$$

This last form is taken as the definition.

Definition 4.2 — Independent events. Two events A and B are **independent** if and only if (iff)

$$P(A \cap B) = P(A)\,P(B).$$

When $P(A) \neq 0$ and $P(B) \neq 0$ this is equivalent to $P(A \mid B) = P(A)$, and to $P(B \mid A) = P(B)$. Events that are not independent are **dependent**.

Note. The original defines independence through $P(A \mid B) = P(A)$. The product form is preferred for two reasons: it still makes sense when $P(B) = 0$, where $P(A \mid B)$ is undefined; and it is symmetric — it visibly says the same thing about A and B, so there is no need to check both directions.

To **use** independence, multiply. To **test** for it, compute $P(A \cap B)$ and $P(A)P(B)$ separately and compare.

Example 5. A code consists of a digit chosen from 0 to 9 followed by a letter of the alphabet. What is the probability that the code is 9Z?

Solution. The choice of digit and the choice of letter are independent. Since

$$P(9) = \frac{1}{10} \quad \text{and} \quad P(Z) = \frac{1}{26},$$

$$P(9Z) = P(9)\,P(Z) = \frac{1}{10} \times \frac{1}{26} = \frac{1}{260}.$$

Counting agrees: there are $10 \times 26 = 260$ equally likely codes, and 9Z is one of them — the multiplication principle of Chapter 1 **is** independence.

Example 6. For events A and B, let $P(A) = 0.1$, $P(B) = K$ and $P(A \cup B) = 0.6$. Find K so that A and B are independent.

Solution. For independent events $P(A \cap B) = P(A)\,P(B) = 0.1K$. Also, by the addition rule,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 0.6 = 0.1 + K - P(A)P(B).$$

Thus

$$0.6 = 0.1 + K - 0.1K \Rightarrow 0.5 = 0.9K \Rightarrow K = \frac{5}{9}.$$

> **Note.** Compare Example 9 of Chapter 3, where the events were **mutually exclusive** and the subtracted term was 0. The setup is identical; only the value of $P(A \cap B)$ changes — 0 for mutually exclusive, $P(A)P(B)$ for independent.

Example 7. A coin is flipped and a die is rolled. Find the probability of getting a head on the coin and a 2 on the die.

Solution. The coin and the die do not affect each other, so "head" and "2" are independent:

$$P(\text{head and } 2) = P(\text{head}) \cdot P(2) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}.$$

Check by listing: the sample space has $2 \times 6 = 12$ equally likely outcomes $(H,1), ..., (T,6)$, and $(H,2)$ is one of them.

In Examples 5 and 7 independence was **assumed**, from the physics of the experiment: separate coins, dice and choices do not influence each other. The next example **tests** for it, from the numbers — and the answer is not obvious in advance.

Example 8. A coin is tossed three times. Let A be the event that a head appears on the second toss and B the event that a tail appears exactly once. Are A and B independent?

Solution.

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\},$$

eight equally likely outcomes, and

$$A = \{HHH, HHT, THH, THT\}, \quad B = \{HHT, HTH, THH\}.$$

Hence

$$P(A) = \frac{4}{8} = \frac{1}{2}, \quad P(B) = \frac{3}{8} \quad \Rightarrow \quad P(A)\,P(B) = \frac{3}{16}.$$

On the other hand,

$$A \cap B = \{HHT, THH\} \Rightarrow P(A \cap B) = \frac{2}{8} = \frac{1}{4} \neq \frac{3}{16} = P(A)\,P(B).$$

So A and B are **not** independent.

> **Note.** **In words.** $P(B \mid A) = (1/4)/(1/2) = 1/2$, up from $P(B) = 3/8$. Knowing the middle toss is a head makes "exactly one tail" more likely: the one tail must then fall on toss 1 or toss 3, and two of the four outcomes in A do that.

> **Watch out.** **Independent is not the same as mutually exclusive** — they are close to opposites. If A and B are mutually exclusive and both possible, then knowing A occurred tells you for certain that B did not: $P(B \mid A) = 0 \neq P(B)$. In symbols, $P(A \cap B) = 0$ but $P(A)P(B) > 0$. Mutually exclusive is a statement about **sets** (no shared outcomes); independent is a statement about **probabilities** (no information flows), and it cannot be seen in a Venn diagram.

Exercise set 2

2.1 A and B are independent with $P(A) = 0.3$ and $P(B) = 0.4$. Find $P(A \cap B)$, $P(A \cup B)$ and $P(A \mid B)$.

2.2 Decide whether A and B are independent: (a) $P(A) = 0.2$, $P(B) = 0.5$, $P(A \cap B) = 0.1$; (b) $P(A) = 0.3$, $P(B) = 0.4$, $P(A \cup B) = 0.6$.

2.3 For the codes of Example 5, find the probability that the digit is even and the letter is a vowel (A, E, I, O, U).

2.4 A and B are mutually exclusive with $P(A) = 0.3$ and $P(B) = 0.4$. Can they be independent? Explain.

2.5 A fair die is rolled twice. Find the probability of two sixes, of no six, and of at least one six.

2.6 $P(A) = 0.4$ and $P(A \cup B) = 0.7$. Find $P(B)$ if (a) A and B are mutually exclusive; (b) A and B are independent.

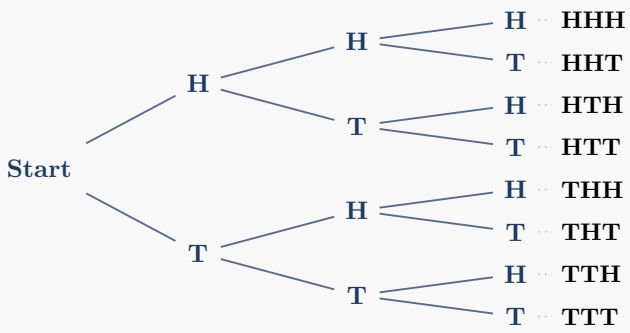
4 Tree diagrams and listing outcomes

A tree is also the quickest way to **list** a sample space built in stages, with no probabilities on the branches at all. Each path from Start to a leaf is one outcome.

Example 9. Draw a tree diagram to show the results of tossing a coin three times.

1. What is the probability of getting 2 heads?
2. What is the probability of getting at least 2 heads?
3. What is the probability of getting at most 2 heads?

Solution.



Reading the leaves from top to bottom,

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\},$$

eight equally likely outcomes. Translate each question first:

(a) means **exactly** 2 heads;
(b) means exactly 2 heads **or** 3 heads;
(c) means exactly 0 heads, 1 head **or** 2 heads.

Count the outcomes by number of heads:

Number of heads	**0**	**1**	**2**	**3**	**Total**
Number of outcomes	1	3	3	1	8

The events "exactly k heads" are mutually exclusive, so their probabilities add:

(a) $P(2H) = 3/8$.

(b) $P(2H \text{ or } 3H) = P(2H) + P(3H) = \frac{3}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$.

(c) $P(0H, 1H \text{ or } 2H) = P(0H) + P(1H) + P(2H) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}$.

Note. Part (c) is faster by the complement: "at most 2 heads" fails only for HHH, so the answer is $1 - 1/8 = 7/8$. And the tree connects to §3: the tosses are independent, so every branch carries $1/2$ and every path has probability $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$ — which is **why** the eight outcomes are equally likely.

5 Two-way tables

When data are classified by two characteristics at once, the counts go in a two-way (contingency) table. A conditional probability then means: **restrict to one row or column, and take a share of it.**

Example 10. One hundred people were classified according to gender (male or female) as well as according to colour blindness, with the following results:

	Colour blind (A)	**Not colour blind**	**Total**
Male (B)	2	58	60
Female	1	39	40
Total	3	97	100

One person is chosen at random. Let A be the event that the person is colour blind and B the event that the person is male.

1. Find the probability that the chosen person is colour blind, given that they are male.
2. Find the probability that the chosen person is male, given that they are colour blind.

Solution.

1. Restrict to the **Male row** — 60 people, 2 of them colour blind:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{2/100}{60/100} = \frac{2}{60} = \frac{1}{30}.$$

2. Restrict to the **colour-blind column** — 3 people, 2 of them male:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{2/100}{3/100} = \frac{2}{3}.$$

Note. **Correction.** The original handout gives the answer to part 2 as 2. A probability can never exceed 1 (Theorem 1 of Chapter 3) — the hundredths cancel to leave $2/3$. A quick $0 \le P \le 1$ check catches this kind of slip.

Watch out. $P(A \mid B)$ and $P(B \mid A)$ are different numbers — here $1/30$ against $2/3$. Only 1 in 30 men is colour blind, yet 2 in 3 colour-blind people are men. The numerator $P(A \cap B)$ is the same in both; what changes is **what you divide by** — the condition, the thing after the bar. Confusing the two is the most common error in this chapter, and the whole subject of the next one (Bayes' theorem).

5.1 The method, in three lines

Step	What you write	Where it goes wrong
1. Spot the condition	"given", "if you know", "the first was …" — the event **after** the bar	Writing $P(A \mid B)$ when the question asked for $P(B \mid A)$
2. Pick the tool	Reduced sample space or $P(A \cap B)/P(B)$; trees for stages; multiply only for independent events	Multiplying $P(A)P(B)$ for events that are **not** independent
3. Compute, then check	The number, $0 \le P \le 1$; tree branches and leaves each sum to 1	A conditional "probability" of 2, left unnoticed

Exercise set 3

3.1 A coin is tossed three times (use the tree of Example 9). Find the probability of exactly one head, of at least one head, and of a head on the first toss given that exactly two heads appear.

3.2 In the setting of Example 8, let C be the event that all three tosses show the same face. Are A (head on the second toss) and C independent?

3.3 Using the table of Example 10, find the probability that the person is female given that they are colour blind, and colour blind given that they are female. Are A and B independent?

3.4 If A and B are independent events, prove that (a) A and B^c, (b) A^c and B, (c) A^c and B^c are independent events.

3.5 Two archers shoot at a target independently. The first hits with probability 0.7, the second with probability 0.6. Find the probability that both hit, that the target is hit at least once, and that exactly one of them hits.

6 Answers

6.1 Set 1 — conditional probability (§1–§2)

1.1 Given "greater than 3" the sample space is $\{4,5,6\}$, so $P(6 \mid \text{greater than } 3) = 1/3$. Given "even" it is $\{2,4,6\}$, and two of those exceed 3: $P = 2/3$.

1.2 $P(A \mid B) = 0.2/0.5 = 0.4$; $P(B \mid A) = 0.2/0.6 = 1/3$; $P(A \mid B^c) = P(A \cap B^c)/P(B^c) = (0.6 - 0.2)/(1 - 0.5) = 0.8$.

1.3 Both hearts: $13/52 \cdot 12/51 = 1/17$. Heart then spade: $13/52 \cdot 13/51 = 13/204$ — the first heart takes no spade away.

1.4 Both red: $3/7 \cdot 2/6 = 1/7$. Different colours: red–blue plus blue–red, $3/7 \cdot 4/6 + 4/7 \cdot 3/6 = 4/7$. Check: both blue is $4/7 \cdot 3/6 = 2/7$, and $1/7 + 4/7 + 2/7 = 1$. ✓

1.5 Given the first die is 3, the total is 8 only if the second is 5: $P = 1/6$. Given total 8, the reduced space is $\{(2,6),(3,5),(4,4),(5,3),(6,2)\}$, so $P(\text{first is } 3 \mid \text{total } 8) = 1/5$. Same intersection, different conditions, different answers.

1.6 $4/52 \cdot 3/51 \cdot 2/50 = 24/132,600 = 1/5525$.

6.2 Set 2 — independence (§3)

2.1 $P(A \cap B) = 0.3 \times 0.4 = 0.12$; $P(A \cup B) = 0.3 + 0.4 - 0.12 = 0.58$; $P(A \mid B) = P(A) = 0.3$.

2.2 (a) $P(A)P(B) = 0.1 = P(A \cap B)$: independent. (b) $P(A \cap B) = 0.3 + 0.4 - 0.6 = 0.1$, but $P(A)P(B) = 0.12$: dependent.

2.3 Independent choices: $5/10 \cdot 5/26 = 25/260 = 5/52$.

2.4 No. Mutually exclusive gives $P(A \cap B) = 0$, while independence would need $P(A \cap B) = 0.3 \times 0.4 = 0.12$. Two mutually exclusive events can only be independent if one of them has probability 0.

2.5 Two sixes: $1/6 \cdot 1/6 = 1/36$. No six: $5/6 \cdot 5/6 = 25/36$. At least one six: $1 - 25/36 = 11/36$.

2.6 (a) $0.7 = 0.4 + P(B)$, so $P(B) = 0.3$. (b) $0.7 = 0.4 + p - 0.4p$, so $0.3 = 0.6p$ and $P(B) = 0.5$.

6.3 Set 3 — trees, tables and proofs (§4–§5)

3.1 Exactly one head: $\{HTT, THT, TTH\}$, 3/8. At least one head: $1 - P(TTT) = 7/8$. Given exactly two heads, the space is $\{HHT, HTH, THH\}$, and two of the three start with H: 2/3.

3.2 $C = \{HHH, TTT\}$, $P(C) = 1/4$; $A \cap C = \{HHH\}$, $P(A \cap C) = 1/8$. Since $P(A)P(C) = 1/2 \cdot 1/4 = 1/8$, they **are** independent — even though the outcomes overlap and nothing about the coin suggests it in advance.

3.3 $P(\text{female} \mid A) = 1/3$; $P(A \mid \text{female}) = 1/40$. $P(A \cap B) = 0.02$, while $P(A)P(B) = 0.03 \times 0.60 = 0.018$: not independent (colour blindness is more common among men: $1/30 > 1/40$).

3.4 (a) A is the disjoint union of $A \cap B$ and $A \cap B^c$, so by Corollary 1 of Chapter 3 and independence,

$$P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)\,P(B^c).$$

(b) The same argument with the roles of A and B exchanged. (c) By (a), A and B^c are independent; applying (b) to that pair (with B^c in place of B) shows A^c and B^c are independent. Directly, with De Morgan:

$$\begin{aligned} P(A^c \cap B^c) &= 1 - P(A \cup B) = 1 - P(A) - P(B) + P(A)P(B) \\ &= (1 - P(A))(1 - P(B)) = P(A^c)\,P(B^c). \end{aligned}$$

3.5 Both: $0.7 \times 0.6 = 0.42$. At least once: $1 - P(\text{both miss}) = 1 - 0.3 \times 0.4 = 0.88$ (the misses are independent too, by question 4). Exactly one: $0.7 \times 0.4 + 0.3 \times 0.6 = 0.46$. Check: $0.88 - 0.42 = 0.46$. ✓